# Ziping Xu

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## Current Position

2025 – Now **Assistant Professor, University of North Carolina, Chapel Hill**, *(From July 1$^{st}$, 2025)*.
School of Data and Information Sciences

## Education

2018–2023 **Ph.D. in Statistics, University of Michigan**, Ann Arbor, MI.
Advisor: Prof. Ambuj Tewari
Thesis: On the Benefits of Multitask Learning: A Perspective Based on Task Diversity

2014–2018 **B.S. in Data Science, Peking University**, Beijing, China.
Advisor: Prof. Song Xi Chen
Thesis: Meteorological Change and Impacts on Air Pollution: Results from North China

## Research Interest

Statistical foundations of sequential decision-making with scarce, adaptively collected human data: reinforcement learning and bandit theory across tasks and deployments; valid causal inference after adaptive experimentation; and calibrated generative models (diffusion and language-model simulators, digital twins) for designing interventions before they reach people. Applications in mobile health and digital interventions, including algorithms deployed in clinical trials.

## Publications and Preprints

(* denotes equal contribution or alphabetical order)

2026 **Ziping Xu***, Yuyi Chang*, Chenshun Ni, Nithin Sugavanam, Asim H. Gazi, Predrag Klasnja, Emre Ertin, and Susan A. Murphy. A diffusion-model subpopulation digital twin for mobile health deployment: A case study on the HeartSteps intervention. *arXiv preprint arXiv:2607.21403*, 2026.

2026 Kevin Tan and **Ziping Xu**. From POMDP theory to deep RL with particle filters. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. arXiv:2610.27575.

2026 Yongyi Guo*, Zifan Xu*, **Ziping Xu***, and Kelly W. Zhang*. Uncertainty-aware selection of online algorithms with simulator ensembles. *arXiv preprint arXiv:2609.32170*, 2026.

2026 Yongyi Guo* and **Ziping Xu***. Statistical inference for misspecified contextual bandits. *arXiv preprint*, 2026.

2026 S. Ghosh, B. T. Gullapalli, Daiqi Gao, Asim H. Gazi, Anna L. Trella, **Ziping Xu**, Kelly Zhang, et al. Reproducible workflow for online artificial intelligence in digital health. *Philosophical Transactions of the Royal Society A*, 2026.

2026 Asim H. Gazi, Yongyi Guo, Daiqi Gao, **Ziping Xu**, Kelly W. Zhang, and Susan A. Murphy. Reinforcement learning in the real world: A survey of statistical challenges and future directions. *arXiv preprint arXiv:2601.15353*, 2026.

2026 Asim H. Gazi, S. Ghosh, Yongyi Guo, Daiqi Gao, **Ziping Xu**, Inbal Nahum-Shani, and Susan Murphy. Key concepts in online learning and decision making for personalizing just-in-time adaptive interventions. *Annals of Behavioral Medicine*, 2026.

2026 Asim H. Gazi, Daiqi Gao, **Ziping Xu**, Predrag Klasnja, Jin-Oh Hahn, and Susan A. Murphy. Overcoming data impoverishment, a threat to the continuity of digital twins in digital health. *Progress in Biomedical Engineering*, 2026.

2026 Asim H. Gazi, Daiqi Gao, S. Ghosh, **Ziping Xu**, Anna L. Trella, Predrag Klasnja, and Susan A. Murphy. Digital twins for just-in-time adaptive interventions (JITAIs): Framework for optimizing and continually improving JITAIs. *Journal of Medical Internet Research*, 2026.

2026 Daiqi Gao, **Ziping Xu**, Aseel Rawashdeh, Predrag Klasnja, and Susan Murphy. Active measuring in reinforcement learning with delayed negative effects. In *Proceedings of the International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026.

2025 Lutong Zou, **Ziping Xu**, Daiqi Gao, and Susan Murphy. Causal directed acyclic graph-informed reward design. In *The Multi-disciplinary Conference on Reinforcement Learning and Decision Making (RLDM)*, 2025.

2025 Anna L Trella, Walter Dempsey, Asim H Gazi, **Ziping Xu**, Finale Doshi-Velez, and Susan A Murphy. Non-stationary latent auto-regressive bandits. *Reinforcement Learning Journal*, 2025.

2025 **Ziping Xu**, Hinal Jajal, Sung Won Choi, Inbal Nahum-Shani, Guy Shani, Alexandra Psihogios, Pei-Yao Hung, and Susan Murphy. Reinforcement learning on aya dyads to enhance medication adherence. In *International Conference on Artificial Intelligence in Medicine (AIME)*, 2025.

2024 **Ziping Xu**, Iris Yan, and Susan Murphy. An adaptation of RLSVI with explicit action sampling probabilities. In *RLC 2024 Deployable RL Workshop*, 2024.

2024 **Ziping Xu**, Zifan Xu, Runxuan Jiang, Peter Stone, and Ambuj Tewari. Sample efficient myopic exploration through multitask Reinforcement Learning with diverse tasks. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.

2024 **Ziping Xu**, Kelly Zhang, and Susan Murphy. The fallacy of minimizing cumulative regret in the sequential task setting. *arXiv*, 2024.

2024 Kevin Tan* and **Ziping Xu***. A natural extension to online algorithms for hybrid RL with limited coverage. *Reinforcement Learning Journal*, volume 1, 2024.

2024 Yongyi Guo*, **Ziping Xu***, and Susan Murphy. Online learning in bandits with predicted context. *In Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.

2024 Quanfu Fan, Yilai Li, Yuguang Yao, John Cohn, Sijia Liu, **Ziping Xu**, Seychelle Vos, and Michael Cianfrocco. CryoRL: Reinforcement Learning enables efficient cryo-EM data collection. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 7892–7902, 2024.

2023 **Ziping Xu**, Quanfu Fan, Yilai Li, Emma Rose Lee, John Maxwell Cohn, Ambuj Tewari, Seychelle M Vos, and Michael Cianfrocco. Coupling semi-supervised learning with reinforcement learning for better decision making—an application to cryo-EM data collection. In *NeurIPS 2023 AI for Science Workshop*, 2023.

2022 **Ziping Xu** and Ambuj Tewari. On the statistical benefits of curriculum learning. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*. PMLR, 2022.

2022 **Ziping Xu**, Eunjae Shim, Ambuj Tewari, and Paul Zimmerman. Adaptive sampling for discovery. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, 2022.

2022 Hao Sun, **Ziping Xu**, Taiyi Wang, Meng Fang, and Bolei Zhou. Supervised q-learning for continuous control. In *Deep Reinforcement Learning Workshop NeurIPS 2022*, 2022.

2022 Hao Sun, **Ziping Xu**, Zhenghao Peng, Meng Fang, Bo Dai, and Bolei Zhou. Mopa: a minimalist off-policy approach to safe-RL. In *Deep Reinforcement Learning Workshop NeurIPS 2022*, 2022.

2022 Eunjae Shim, Joshua A Kammeraad, **Ziping Xu**, Ambuj Tewari, Tim Cernak, and Paul M Zimmerman. Predicting reaction conditions from limited data through active transfer learning. *Chemical Science*. Royal Society of Chemistry, 2022.

2022 Aditya Modi, **Ziping Xu**, Mohamad KS Faradonbeh, and Ambuj Tewari. Big control actions help multitask learning of unstable linear systems. *In ICML 2022 Complex feedback in online learning Workshop*, 2022.

2022 Yilai Li, Quanfu Fan, **Ziping Xu**, Emma Rose Lee, John Cohn, Veronique Demers, Ja Young Lee, Lucy Yip, Michael A. Cianfrocco, and Seychelle M. Vos. Optimized path planning surpasses human efficiency in cryo-EM imaging. *eLife*. Cold Spring Harbor Laboratory, 2022.

2021 **Ziping Xu** and Ambuj Tewari. Representation learning beyond linear prediction functions. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, 2021.

2021 **Ziping Xu**, Amirhossein Meisami, and Ambuj Tewari. Decision making problems with funnel structure: A multi-task learning approach with application to email marketing campaigns. In *In Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 127–135. PMLR, 2021.

2021 Yangyi Lu*, **Ziping Xu***, and Ambuj Tewari. Bandit algorithms for precision medicine. *arXiv preprint arXiv:2108.04782*, 2021.

2020 **Ziping Xu** and Ambuj Tewari. Reinforcement learning in factored MDPs: Oracle-efficient algorithms and tighter regret bounds for the non-episodic setting. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, 2020.

2020 **Ziping Xu**, Song Xi Chen, and Xiaoqing Wu. Meteorological change and impacts on air pollution: Results from north China. *Journal of Geophysical Research: Atmospheres*, volume 125, page e2020JD032423. Wiley Online Library, 2020.

2020 Tarun Gogineni, **Ziping Xu**, Exequiel Punzalan, Runxuan Jiang, Joshua Kammeraad, Ambuj Tewari, and Paul Zimmerman. Torsionnet: A Reinforcement Learning approach to sequential conformer search. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

2019 **Ziping Xu** and Ambuj Tewari. Worst-case regret bound for perturbation based exploration. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

2018 Wenling Liu, **Ziping Xu**, and Tianan Yang. Health effects of air pollution in china. *International journal of environmental research and public health*, volume 15, page 1471. MDPI, 2018.

2017 Shuyi Zhang, Bin Guo, Anlan Dong, Jing He, **Ziping Xu**, and Song Xi Chen. Cautionary tales on air-quality improvement in Beijing. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, volume 473, page 20170457. The Royal Society Publishing, 2017.

## Fellowships & Awards

2026 ***NSF Travel Award*** (2026 ICSA Applied Statistics Symposium)

2026 ***NSF Travel Award*** (9th International Workshop in Sequential Methodologies, IWSM)

2025 ***Spotlight Presentation*** (top 6%, 20 of 339 accepted abstracts), 6th Multidisciplinary Conference on Reinforcement Learning and Decision Making (RLDM)

2024 ***Doctoral Consortium*** (selected participant) and Travel Award, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)

2023 ***Departmental Outstanding Dissertation Award*** (Awarded for outstanding dissertation writing)

2022 ***Rackham Predoctoral Fellowship*** (Awarded to outstanding doctoral students)

2021 ***Spotlight Presentation*** (top 3% of 9,122 submissions), Conference on Neural Information Processing Systems (NeurIPS), for "Representation Learning Beyond Linear Prediction Functions"

2020 ***Outstanding Graduate Student Instructor Awards*** (Awarded to outstanding instructors)

2016 ***Lee Wai Wing Scholarship*** (Awarded to outstanding first year students)

## Funding

2026–2029 Principal Investigator (UNC–Chapel Hill component), National Science Foundation, Division of Mathematical Sciences, FDT-BioTech program (NSF 24-561). "Collaborative Research: FDT-BioTech: Mechanistic-Neural Foundations for Digital Twins of Parkinson's Disease Symptom Dynamics Under Levodopa Treatment." Award DMS-2625746, $400,000 ($800,000 total collaborative), Sept. 2026–Aug. 2029. With co-funding from the National Institutes of Health.

## Invited Talks and Presentations

### Conference and Workshop Talks

Dec. 2026 "Subpopulation Digital Twins for Mobile Health Deployment" (invited, scheduled), ASA Mobile and Wearable Data Science Interest Group Webinar Series

June 2026 "A Diffusion-Model Subpopulation Digital Twin for Mobile Health Deployment: A Case Study on the HeartSteps Intervention", 2026 ICSA Applied Statistics Symposium

June 2026 "The Fallacy of Minimizing Cumulative Regret in the Sequential Task Setting", 9th International Workshop in Sequential Methodologies (IWSM), American University, Washington, DC

Oct. 2025 Invited talk, INFORMS Annual Meeting (Health Applications Society session)

Aug. 2025 Invited discussant, AOAS Discussion Session, Joint Statistical Meetings (JSM) 2025

June 2025 Invited session talk, New England Statistics Symposium (NESS) 2025

June 2025 Spotlight presentation, 6th Multidisciplinary Conference on Reinforcement Learning and Decision Making (RLDM), Dublin

Oct. 2025 Co-organizer and speaker, Special Session, IEEE International Conference on Biomedical and Health Informatics (BHI) 2025

Oct. 2024 Invited talk, INFORMS Annual Meeting 2024

Aug. 2024 Contributed talk, Joint Statistical Meetings (JSM) 2024, Portland

June 2023 Invited talk, 2023 ICSA Applied Statistics Symposium, Ann Arbor

### Invited Seminars and Colloquia

2026 Statistics Seminar, Rutgers University

2025 Department of Statistical Sciences, University of Toronto

2025 Department of Statistics, University of California, Riverside

2025 Department of Biostatistics, University of North Carolina at Chapel Hill

2025 School of Data Science and Society, University of North Carolina at Chapel Hill

2025 Department of Biostatistics, University of Michigan

2024 University of Sydney

2024 AI & Healthcare Seminar Series, Mila / McGill University

2024 mDOT Center Webinar, NIH mHealth Center of Excellence

2022 Department of Statistics, University of Florida

2022 University of Massachusetts Amherst

## Teaching Experiences

### Instructor

Spring 2027 ***DATA 215*** (scheduled; core course for the B.S. in Data Science) University of North Carolina at Chapel Hill

Spring 2026 ***DATA 521, Foundations of AI*** University of North Carolina at Chapel Hill

### Teaching Fellow

Winter 2024 ***STATS 234, Sequential Decision Making*** Harvard University

Graduate Student Instructor

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| Winter 2020 | ***STATS 503, Statistical Learning II*** | University of Michigan |
| Fall 2019 | ***STATS 415, Introduction to Data Mining*** | University of Michigan |
| Winter 2019 | ***STATS 250, Introduction to Statistics and Data Analysis*** | University of Michigan |
| Fall 2018 | ***STATS 425, Introduction to Probability*** | University of Michigan |

## Mentoring Experience

2024-Now ***Lutong Zou***
Topic: Reward Design through Causal DAG

2023-2024 ***Iris Yan***
Topic: Approximate Sampling Probability for Online Decision Making

2022 ***Zhiyu Yuan*** (Next Step: CMU, MS in Computer Engineer)
Topic: Curriculum Learning for Reinforcement Learning

2022 ***Ruanxuan Jiang*** (Next Step: Software Engineer Citadel)
Topic: Theory of Multitask Reinforcement Learning

2020–2021 ***Anh Tuan (Alan) Tran*** (Next Step: University of Toronto, MS in Computer Science)
Topic: Sample Efficient Reinforcement Learning

## Professional Experience

2023 – 2025 Postdoctoral Research Fellow at **Department of Statistics, Harvard University**

2025 – now Associate (unpaid appointment), **Department of Statistics, Harvard University**
Advisor: Dr. Susan A. Murphy, Professor, Department of Statistics and Computer Science

Summer 2022 Research Intern at **MIT-IBM Watson AI Lab**
Advisor: Dr. Quanfu Fan

Summer 2020 Research Intern at **Adobe Inc.**
Advisor: Dr. Amirhossein Meisami

## Professional Activities

Organizing Committee

2026 Session Chair, Topic-Contributed Paper Session "Advances in Reinforcement Learning and Adaptive Experimentation", Joint Statistical Meetings (JSM)

2021 Representative Michigan Student Symposium for Interdisciplinary Statistical Sciences

Journal Paper Reviewer

2026 Science
2025 Nature
2026 ACM Transactions on Computing for Healthcare (ACM Health)
2026 Stochastic Systems
2025 Biostatistics
2025 Communications in Statistics – Theory and Methods
2023 Statistical Science
2023 Proceedings of the National Academy of Sciences of the United States of America (PNAS)
2022 Journal of the American Statistical Association (JASA)
2022 Annals of Statistics
2020 Autonomous Agents and Multi-Agent Systems (JAAMAS)

### Conference Paper Reviewer

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|---|---|
| 2022 | International Conference on Artificial Intelligence and Statistics (AISTATS) |
| 2022, 2024, 2026 | International Conference on Learning Representations (ICLR) |
| 2023-2024 | International Conference on Machine Learning (ICML) |
| 2021–2024 | Conference on Neural Information Processing Systems (NeurIPS) |
| 2025–2026 | Association for the Advancement of Artificial Intelligence (AAAI), Program Committee |
| 2025–2026 | Reinforcement Learning Conference (RLC) & Reinforcement Learning Journal (RLJ) |
| 2026 | European Workshop on Reinforcement Learning (EWRL) |

### Grant / Proposal Reviewer

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| 2026 | Swiss National Science Foundation (SNSF) |